

VINOD SITHAMPARAM PILLAI

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SUMMARY

Software Engineer with 3+ years building automated qualification test software, deployment pipelines, and lab equipment control systems for avionics test infrastructure in Linux environments, backed by a Master's in Data Science. Proven track record of eliminating manual test and deployment processes through automation (up to 73% reduction in deployment time), with additional hands-on experience building data-aware and AI/LLM-driven automation workflows using Python, Docker, and cloud infrastructure.

TECHNICAL SKILLS

- **Automation & Scripting:** Python, Bash, SCPI/Telnet Protocol Automation, CI-style Deployment Automation, Test Automation Frameworks
- **Infrastructure & Systems:** Linux Systems Administration, Networking, Docker, Google Cloud Platform (Cloud Run, Cloud SQL, Artifact Registry), GitLab/GitHub Version Control
- **AI & Agentic Automation:** LangGraph, LangChain, scikit-learn, XGBoost, LLM-Integrated Workflow Design
- **Data & Databases:** SQL, PostgreSQL, MySQL, Apache Spark, Databricks
- **Visualization & Tools:** Streamlit, Databricks AI/BI Dashboards

PROFESSIONAL EXPERIENCE

KONTRON ASIA PACIFIC DESIGN

Software Engineer → Software Engineer II, Avionics R&D

Nov 2022 – Mar 2026

- Designed an automated package-installation pipeline to push software patches to production servers, eliminating manual updates across the entire test fleet and cutting total installation time by 73%.
- Automated configuration and data collection for lab test equipment (spectrum analyzers, signal generators, oscilloscopes, RF switches, relay/input-channel boards) via SCPI and Telnet protocols, reducing test modules requiring manual intervention entry from 11 to 4 (64% reduction) and removing the need for an operator to remain stationed at each instrument.
- Maintained and enhanced factory acceptance test (eFAT) automation scripts in Python and Bash for airborne servers in Linux environments – benchmarking peripheral performance against requirements with minimal human intervention and uploading test logs to centralized databases as evidence of functionality prior to customer release.
- Led an architectural shift of the test framework from a Unit Under Test(UUT)-centric to a Test PC (TPC)-centric flow, minimizing required changes on the UUT and centralizing script management on a single entity across releases – improving maintainability and consistency across test cycles
- Contributed to GoTest, a concurrent variant of eFAT running parallel test modules to stress servers to operational limits during environmental testing.
- Proposed and implemented a centralized, version-controlled repository (GitLab) for system software upgrades, standardizing release and deployment workflow across the test-development team – improving code transparency between development and release cycles, enabling faster issue tracking and reduced debugging time, and providing clear verification of whether requirements were met at each release
- Promoted to lead a test-development (FAT) cluster, supervising one developer and owning test strategy and implementation for factory-acceptance workflows.
- Deployed and validated network topologies enabling reliable communication between multiple peripherals and managed switches, and compiled/patched the Linux kernel to meet client-specific requirements.
- Designed and built the frontend and backend of an Energy Management System (MySQL, Python, JavaScript), covering schema design, data storage, and reporting views.
- Partnered directly with customers to diagnose technical issues and deliver tailored solutions on tight timelines.

ACADEMIC & PERSONAL PROJECTS

[US Airline Delay Analytics & Real-Time Data Pipeline](#) — Databricks, Apache Spark, SQL, Amazon S3

- Built an end-to-end lakehouse pipeline on Databricks processing 3.4M+ rows × 110 columns of U.S. domestic flight data (BTS source, Oct 2025–Mar 2026).
- Implemented a Medallion architecture: Bronze for raw ingestion from S3, Silver for schema enforcement, type casting and deduplication, and Gold for business-level aggregates optimized for low-latency reporting.
- Delivered Databricks AI/BI dashboards surfacing carrier-level delay performance, delay-cause attribution, high-risk flight corridors, and ground-taxi efficiency trends for operational stakeholders.

Clinical Triage Control Center — *LangGraph, XGBoost, PostgreSQL, Docker, Streamlit*

- Architected a multi-service containerized pipeline (Streamlit intake portal, LangGraph orchestration engine, PostgreSQL storage) simulating real-time emergency patient triage.
- Designed a stateful LangGraph workflow combining deterministic vitals validation, human-in-the-loop correction, XGBoost + SHAP-based risk prediction, and conditional LLM-driven clinical routing.
- Modelled the relational schema for triage session persistence and audit tracking, and containerized the full stack with Docker Compose for reproducible local deployment.
- Deployed the service to Google Cloud (Cloud Run, Cloud SQL) with images published to Google Artifact Registry for a live demo.

Bank Telemarketing AI Workflow — *scikit-learn, LangGraph, PostgreSQL, Docker, Groq LLM*

- Built a three-service CRM data pipeline (Streamlit dashboard, LangGraph state engine, PostgreSQL) integrating an ensemble classification model with an LLM-based content generator.
- Engineered a feature-transformation layer (derived attributes such as age group and lead type) feeding a conditional routing graph that segments lead and persists outcomes to Postgres for agent use.
- Integrated a Groq-hosted LLM (Qwen3-32B) to generate personalized outreach scripts from model predictions and customer attributes.
- Deployed the service to Google Cloud (Cloud Run, Cloud SQL) with images published to Google Artifact Registry for a live demo.

EDUCATION

- **Master of Data Science**, University of Malaya — GPA 4.00/4.00 *Oct 2025 – Oct 2026 (Est)*
(Part-time -> Full-time from Mar 2026)
- **Bachelor of Mechanical Engineering**, University of Malaya — CGPA 3.60/4.00 *Sept 2018 – July 2022*

LANGUAGES

English, Malay, Tamil

REFERENCES

- **Ahmad Faiz**, Chief Operating Officer (COO) & Manager | +6012 – 490 3665 | Ahmad.Faiz@kontron.com
- **Dr. Aznul Qalid Bin Md Sabri**, Project Supervisor | +6010 – 798 9512 | aznulqalid@um.edu.my